Politecnico di Torino

Collegio di Elettronica, Telecomunicazioni e Fisica

Report for LAB1 of the course Integrated Systems Architecture

Master degree in Electrical Engineering

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CHAPTER 1

Lab 1: design and implementation of a digital filter

1.1 Reference model development

1.1.1 Matlab model

The filter we designed has N = 10 and nb = 11, evaluated from the number of letters in the surnames `Romanisio` and `Garbarino` respectively. This means the fixed point format used in the filter will be 1.10.
The frequency response of the filter is shown in Figure 1.1 this graph is generated by using both the FPU data format, and the quantized fixed point one.Both curves behave similarly up to around 7 kHz, displaying the exact same cutoff frequency f_{-3dB}. After the 7 kHz mark, both graphs display a minimum. The quantized FDT starts falling a bit earlier, and reaches a higher value than the non-quantized FDT. After the negative peak, both curves increase again. The non quantized FDT then converges to a value around -68 dB, while the quantized FDT keeps displaying ripples.

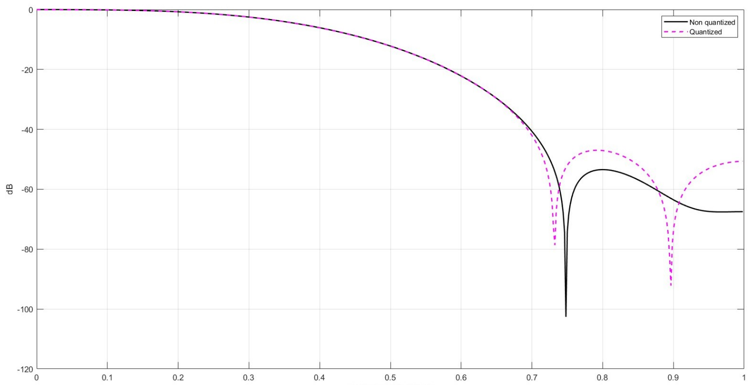

Figure 1.1: Fixed point (quantized) and floating point (not quantized) frequency response of the filter

The filter coefficients have one bit for the integer part and 10 *bits* for the fractional part. They correspond to the integer and fixed point values shown in Table 1.1.

1.1.2 C model and THD

The C model was created from the given sample, by inserting the coefficients in 1.1.1 into `b_i`, setting the coefficients `a_i` to 0, `N` to 10 and `NB` to 11. The `SHAMT` variable was assigned by progressively checking the THD of the generated outputs using the MATLAB script `"distortion.m"`. The THD increase is achieved by shfiting the results of every multiplication by a number of bits equal to `SHAMT`, and then shifting the remaining bits to the right by 3. `SHAMT` was ultimately assigned to 13, which produces outputs of a THD of $\approx -36dB$. Using the first section of the MATLAB code `"Confronto.m"`.

Coefficent	Integer	Floating Point
b_0	-1	$-9.765 \cdot 10^{-4}$
b_1	-13	0.0126
b_2	-26	0.0253
b_3	65	0.0634
b_4	281	0.2744
b_5	407	0.3974
b_6	281	0.2744
b_7	65	0.0634
b_8	-26	-0.0253
b_9	-13	-0.0126
b_{10}	-1	$-9.7656 \cdot 10^{-4}$

Table 1.1: Filter coefficents in integer form and fixed point form

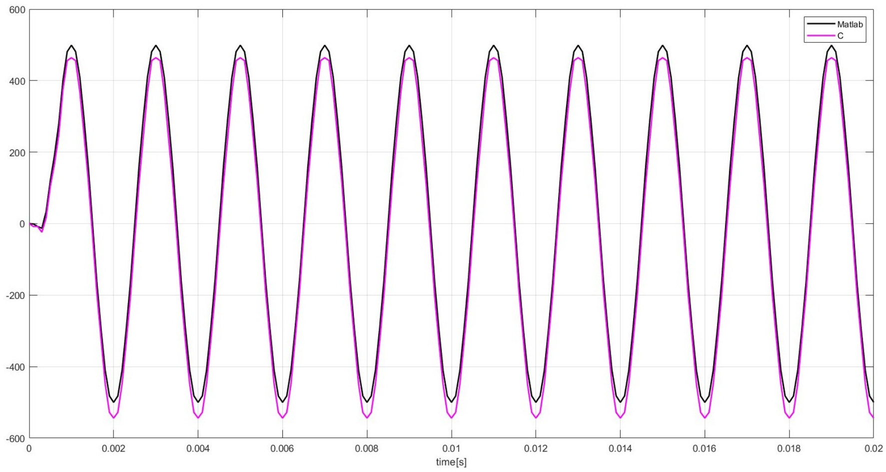

Figure 1.2: Plot of the outputs generated by MatLab vs the ones generated from C

1.1.3 Explanations, comparisons and comments

In 1.2 we can see the comparison between the values produced by Matlab and those produced by the C simulation using the same input samples. Both results behave like a sinusoid with a given frequency $f = 500\ Hz$, but the C outputs display a peak value 35 units lower than the Matlab one. This is caused by the multiplication approximation introduced by the `SHAMT` variable.

1.2 VLSI implementation

1.2.1 Architecture

The architecture of the filter is shown in Figure 1.3 and Figure 1.4 which is a more detailed scheme that shows the approximation method used after the multiplication and the parallelism of the data; the timing diagram is shown in Figure 1.5. The approximation is required to follow properly the C model, and to guarantee the required THD of the outputs. The approximation is implemented by forcing the three LSBs of the result of every multiplication to `"000"`, and to remove the MSB. This mirrors what the `SHAMT` did in 1.1.2, and guarantees the format of the multiplier output is still 1.10.
To better clarify the names used in the timing diagram, a list of arithmetic formulas is shown below; these formulas are not comprehensive of the approximation that the circuit makes after multiplication, but represent only the logical values of the signals $DOUT_i$.

- $DOUT_0 = DIN_0 \cdot b_0$
- $DOUT_1 = DIN_0 \cdot b_1 + DIN_1 \cdot b_0$
- $DOUT_2 = DIN_0 \cdot b_2 + DIN_1 \cdot b_1 + DIN_2 \cdot b_0$
- $DOUT_3 = DIN_0 \cdot b_3 + DIN_1 \cdot b_2 + DIN_2 \cdot b_1 + DIN_3 \cdot b_0$

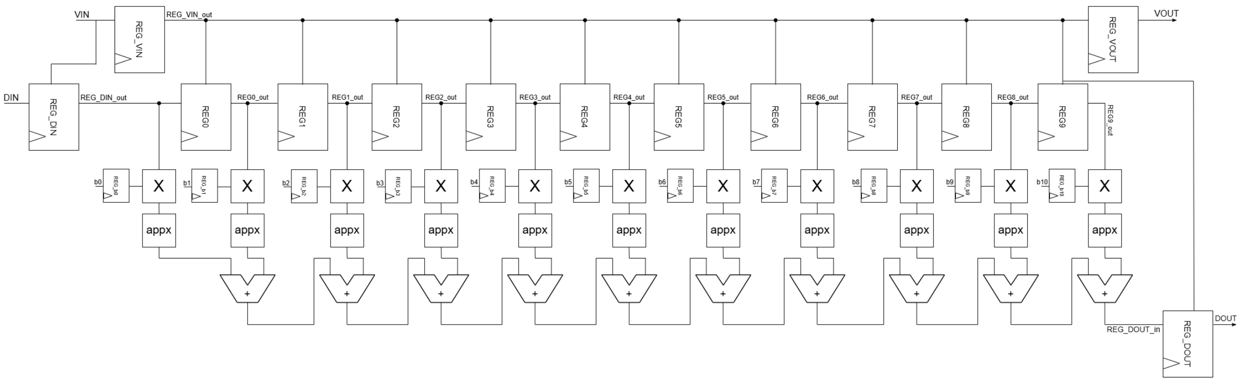

Figure 1.3: FIR architecture

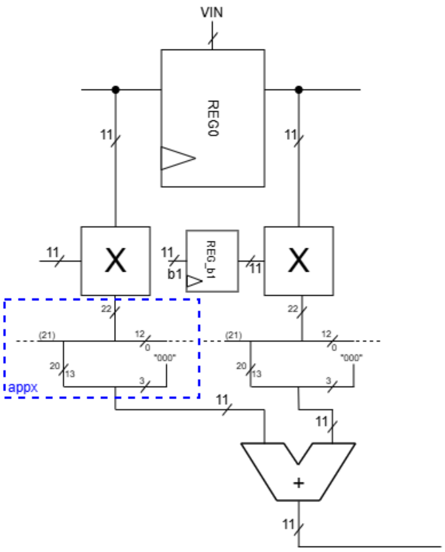

Figure 1.4: FIR architecture in details

- $DOUT_4 = DIN_0 \cdot b_4 + DIN_1 \cdot b_3 + DIN_2 \cdot b_2 + DIN_3 \cdot b_1 + DIN_4 \cdot b_0$
- $DOUT_5 = DIN_0 \cdot b_5 + DIN_1 \cdot b_4 + DIN_2 \cdot b_3 + DIN_3 \cdot b_2 + DIN_4 \cdot b_1 + DIN_5 \cdot b_0$
- $DOUT_6 = DIN_0 \cdot b_6 + DIN_1 \cdot b_5 + DIN_2 \cdot b_4 + DIN_3 \cdot b_3 + DIN_4 \cdot b_2 + DIN_5 \cdot b_1 + DIN_6 \cdot b_0$
- $DOUT_7 = DIN_0 \cdot b_7 + DIN_1 \cdot b_6 + DIN_2 \cdot b_5 + DIN_3 \cdot b_4 + DIN_4 \cdot b_3 + DIN_5 \cdot b_2 + DIN_6 \cdot b_1 + DIN_7 \cdot b_0$
- $DOUT_8 = DIN_0 \cdot b_8 + DIN_1 \cdot b_7 + DIN_2 \cdot b_6 + DIN_3 \cdot b_5 + DIN_4 \cdot b_4 + DIN_5 \cdot b_3 + DIN_6 \cdot b_2 + DIN_7 \cdot b_1 + DIN_8 \cdot b_0$
- $DOUT_9 = DIN_0 \cdot b_9 + DIN_1 \cdot b_8 + DIN_2 \cdot b_7 + DIN_3 \cdot b_6 + DIN_4 \cdot b_5 + DIN_5 \cdot b_4 + DIN_6 \cdot b_3 + DIN_7 \cdot b_2 + DIN_8 \cdot b_1 + DIN_9 \cdot b_0$
- $DOUT_{10} = DIN_0 \cdot b_{10} + DIN_1 \cdot b_9 + DIN_2 \cdot b_8 + DIN_3 \cdot b_7 + DIN_4 \cdot b_6 + DIN_5 \cdot b_5 + DIN_6 \cdot b_4 + DIN_7 \cdot b_3 + DIN_8 \cdot b_2 + DIN_9 \cdot b_1 + DIN_{10} \cdot b_0$

1.2.2 Simulation

To process all the samples, the simulation lasts around 40.6 μs. A snapshot from 0 μs to 3.9 μs is shown in Figure 1.6 to highlight the behavior of the filter when VIN moves from 0 to 1 and vice versa.

The results obtained testing the FIR were perfectly matching with the ones previously saved through the C code, as it's showed in the comparison graph shown in Figure 1.7.

Analyzing the waveforms acquired via the testbench, the $VOUT$ signal seems to be active for a total of 20.55 μs.

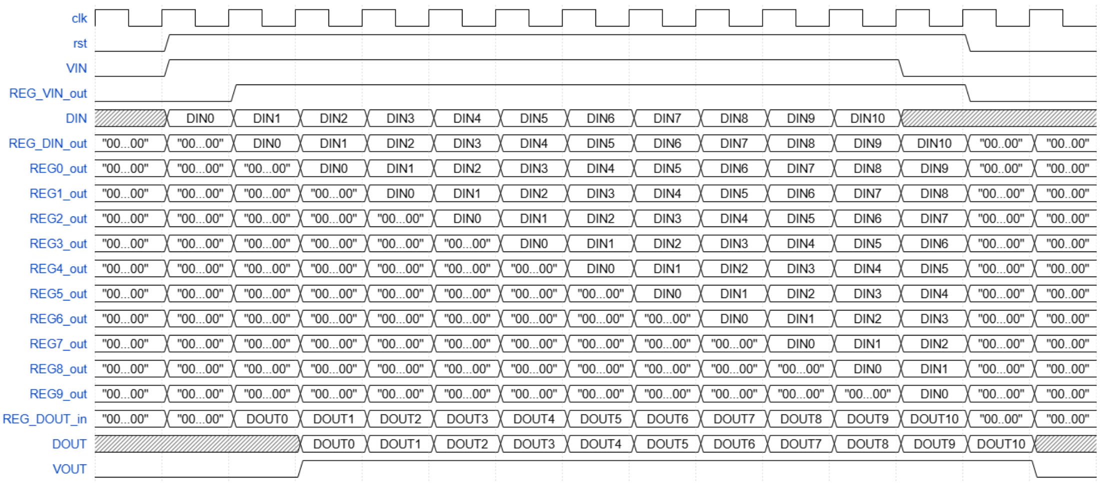

Figure 1.5: Timing diagram

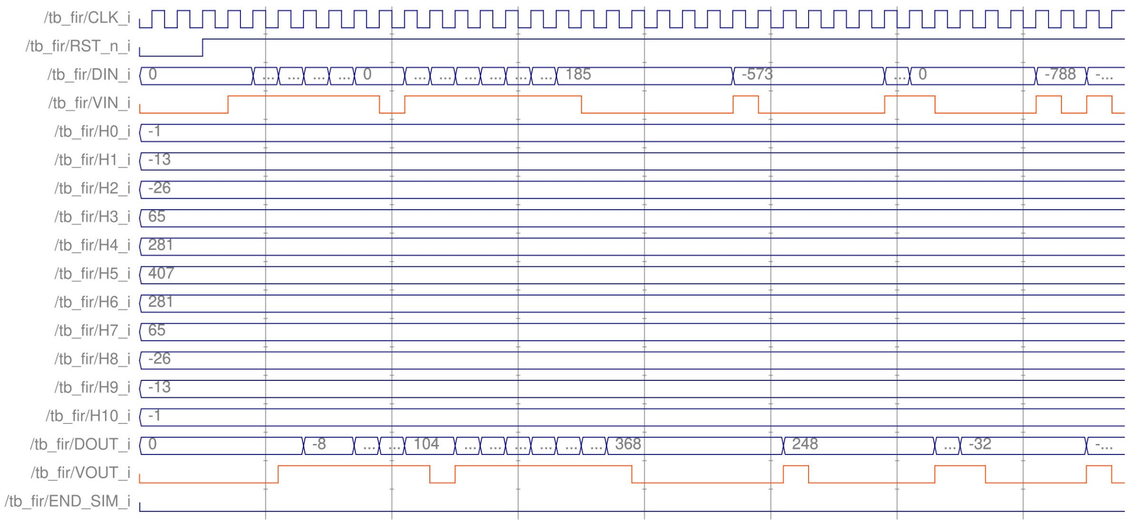

Figure 1.6: FIR's testbench

1.2.3 Logic synthesis

In order to estimate the power,`design_compiler` is used, and the power is estimated using the backannotation method. By imposing a clock period equal to 0 in `design_compiler` it's possible to analyze the slack, and estimate the critical path of the circuit. This is done by changing recursively the clock period constraint until the reported slack turns from negative to 0. The critical clock period is $T_{ck} = 3.8ns$. This value is then inserted in the testbench file `clk_gen.sv` to properly run the backannotation. In `data_maker` the constant T_c is changed to $1ns$ to assure the data generation is quick enough to fit the new clock period. A netlist is generated by `design_compiler` and fed to `QuestaSim` along with `data_maker.vhd`, `data_sink.vhd`, `clock_gen.vhd` and `tb_fir.sv`. By running a simulation `QuestaSim` a `.vcd` file containing the necessary switching activity information is generated. After feeding this `.vcd` file back to `design_compiler` it's possible to properly estimate the power consumption. The backannotation process was applied both with $T_{ck} = 3.8ns$ and $T_{ck} = 7.6ns$. The results of the synthesis are shown in Table 1.2.

Frequency	Area	Power Consumption
$f_M = 263.2\ MHz$	$A = 7540\ \mu m^2$	$P = 1.64\ mW$
$f_M/2 = 131.6\ MHz$	$A = 7525\ \mu m^2$	$P = 897\ \mu W$

Table 1.2: Synthesis results using backannotation

The results of the synthesis with clock gating are shown in Table 1.3, again this synthesis was carried out using backannotation.

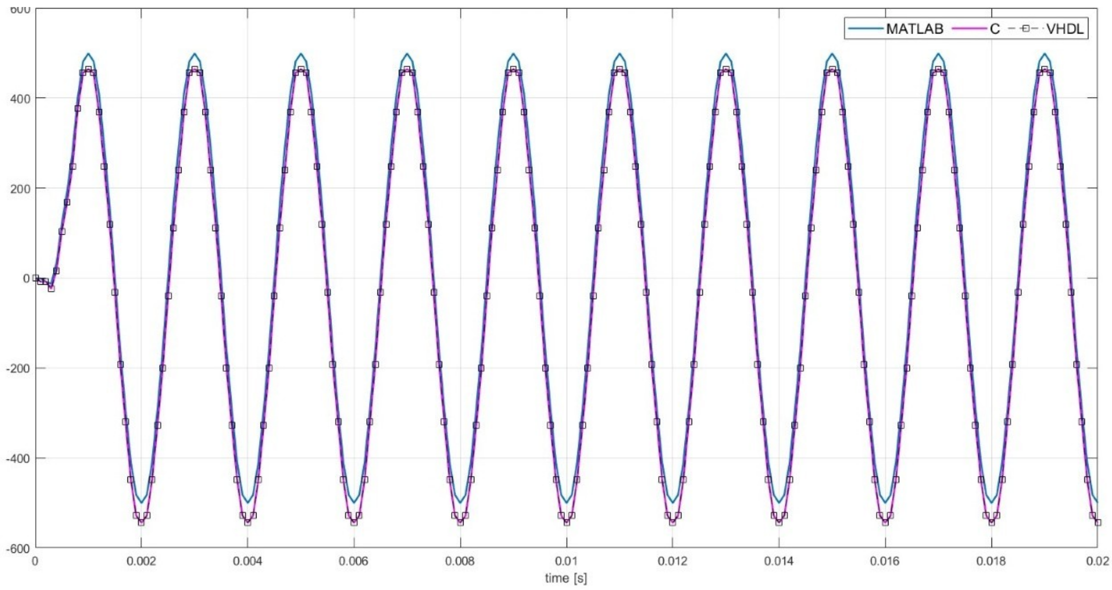

Figure 1.7: Comparison results

Frequency	Area	Power Consumption
$f_M = 263.2\ MHz$	$A =\ 7098\ \mu m^2$	$P =\ 1.471\ mW$
$f_M/2 = 131.6\ MHz$	$A =\ 7089\ \mu m^2$	$P =\ 815\ \mu W$

Table 1.3: Synthesis results using backannotation and clock gating

1.2.4 Place and route

The Place & Route was done using the `Cadence Innovus` environment following the instructions from the assignment. By using the clock period of `7.8 ns` evaluated in 1.2.3 we can observe a minimum slackR (setup slack) of 4.526 ns and a minumum slackF (hold slack) of 0.149 ns. Both slacks are positive, so there is no need to change the clock period as the timing constraints are respected. In order to evaluate the switching activity, a netlist was generated from `Innovus` and fed to `QuestaSim` along with the testbench `tb_fir.sv`. A simulation is run to save the switching activity estimation on a `.vcd` file. The simulation time was measured by evaluating the duration for which the signal `END_SIM_i` remained low obtaining the simulation time of $3,245\ \mu s$. Results of the Place & Route are shown in Table 1.4 and Figure 1.8.

Frequency	Area	Power Consumption
$f_M/2 = 131.6\ MHz$	$A = 7051.9\ \mu m^2$	$P = 732.33\ \mu W$

Table 1.4: Place&Route results

1.2.5 Explanations, comparisons and comments

As can be observed, the clock gating technique - which `Synopsys` automatically implements by disabling the clock in those parts of the circuit where its switching is not required enables a significant reduction in power consumption. It's noticeable a consumption reduction of a factor ≈ 0.9 of all the scale by applying the clock gating to both frequencies of work. Similarly, lowering the switching frequency yields the same effect, since $P = k * f$, halving the frequency reduces the power consumption by a factor ≈ 0.5. Overall, by halving the working frequency and gating the clock, the power consumption is reduced by $\approx 50\%$, the clock gating has a much lower input overall the frequency reduction.
By simulating the circuit under different conditions, both with and without clock gating, and using `Innovus`, the occupied area remains nearly unchanged while the power consumption is reduced by a factor 0.9. The working frequency of the circuit remains unchanged

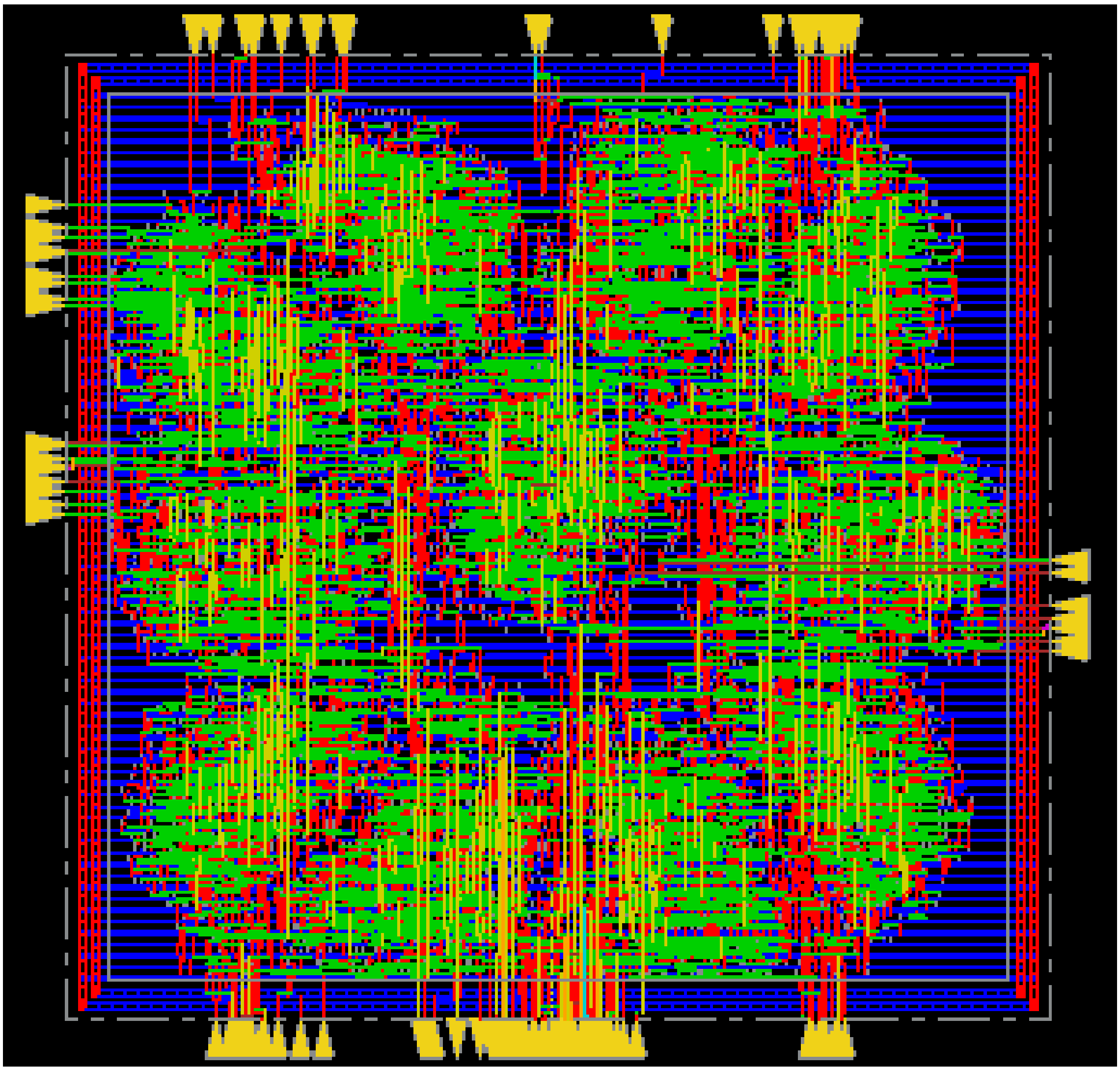

Figure 1.8: Layout of the filter after Place & Route

1.3 Advanced architecture development

1.3.1 Architecture

The architecture implemented is the same filter but with a `Grade 3 unfolding`. The architecture of the filter is shown in Figure 1.6, to derive the architecture the formulas in 1.1 where used, in 1.1 `w` is the number of registers over a connection,`i` is the instance of the unfolding a connection is issued from, `j` is the instance of the unfolding the connection is addressed to, and `P` is the grade of the unfolding (always 3 in our case). Equations 1.1 were used for every connection in the original design. The new design implement the same operation of the folded filter, but now display a parallelism of 3 data. The timing diagram is in Figure 1.9, and refers only to the operations of the registers `R0_i0`,`R1_i0` and `R2_i0` (figure 1.9) for simplicity, the timing of the other 2 register blocks(`Rx_i1` and `Rx_i2`) is equivalent.

$$mod(\frac{i+w_i}{P}) = j \qquad \lfloor\frac{i+w_i}{P}\rfloor = w_{i+1} \tag{1.1}$$

Simulation

To simulate the unfolded filter, the codes `data_maker.vhd`,`data_sink.vhd` and `tb_fir.sv` were modified to support the new parallelism of the Filter. To process all the samples, the simulation lasts around 14.1 μs. A snapshot from 0 ns to 3.9 μs is shown in Figure 1.10 to highlight the behavior of the filter when VIN moves from 0 to 1 and vice versa. The results of the simulation match perfectly with the one previously saved through the C code.

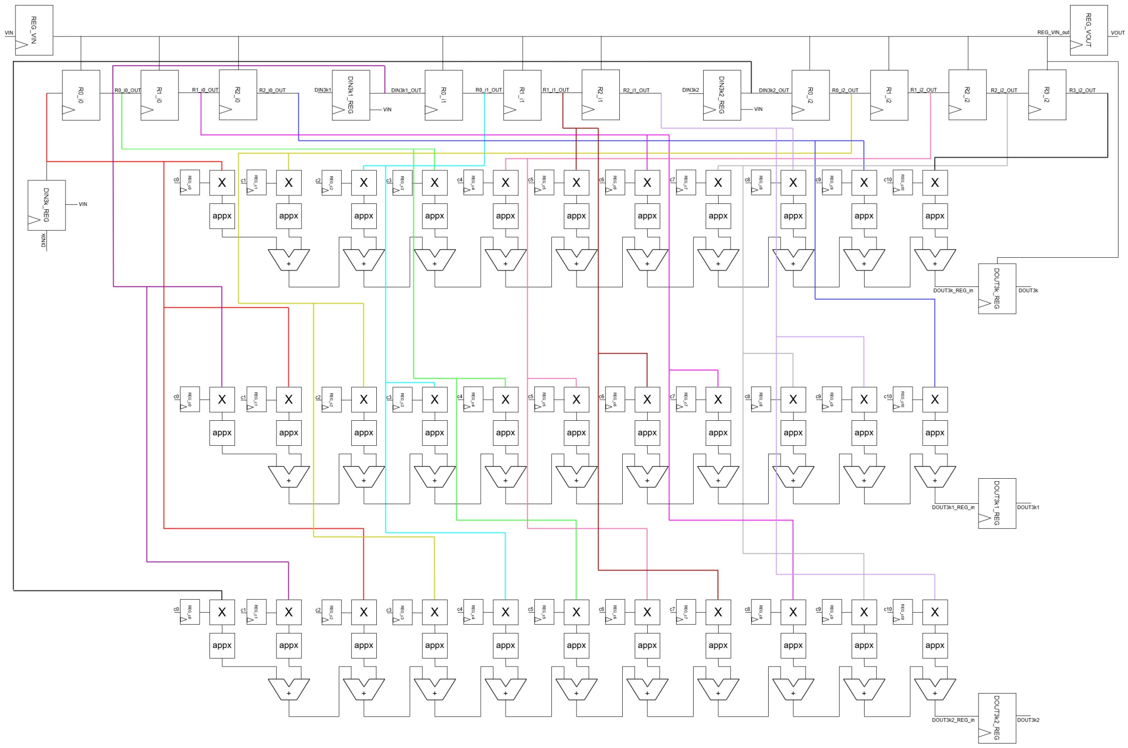

Figure 1.9: FIR realized with unfolding technique

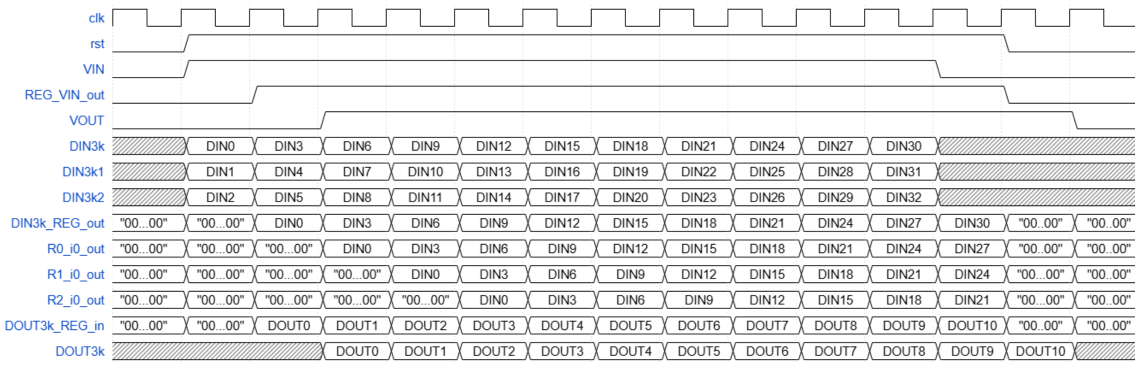

Figure 1.10: Timing of the FIR realized with unfolding technique

1.3.2 Logic synthesis

The synthesis was executed exactly as in 1.2.3. Results of the synthesis are shown in Table 1.5 and in 1.6 with clock gating. From the results we can make some observations. The critical frequency is the same as the folded fir, and so is the halved frequency. The area of the circuit is increased by a factor of ≈ 2.5, which is reasonable given the amount of hardware added by the unfolding. The power consumed is increased by a factor ≈ 3. again this is reasonable by the increase of hardware. The consumption reduction by halving the clock frequency is ≈ 0.7, which is worse than what seen in 1.2.3.The clock gating reduction is still ≈ 0.9 like in 1.2.3

Frequency	Area	Power Consumption
$f_M \;=\; 274\ MHz$	$A \;=\; 18131.9\ \mu m^2$	$P \;=\; 4.40\ mW$
$f_M \;=\; 137\ MHz$	$A \;=\; 17655.5\ \mu m^2$	$P \;=\; 2.98\ mW$

Table 1.5: Unfolded FIR Synthesis results using backannotation

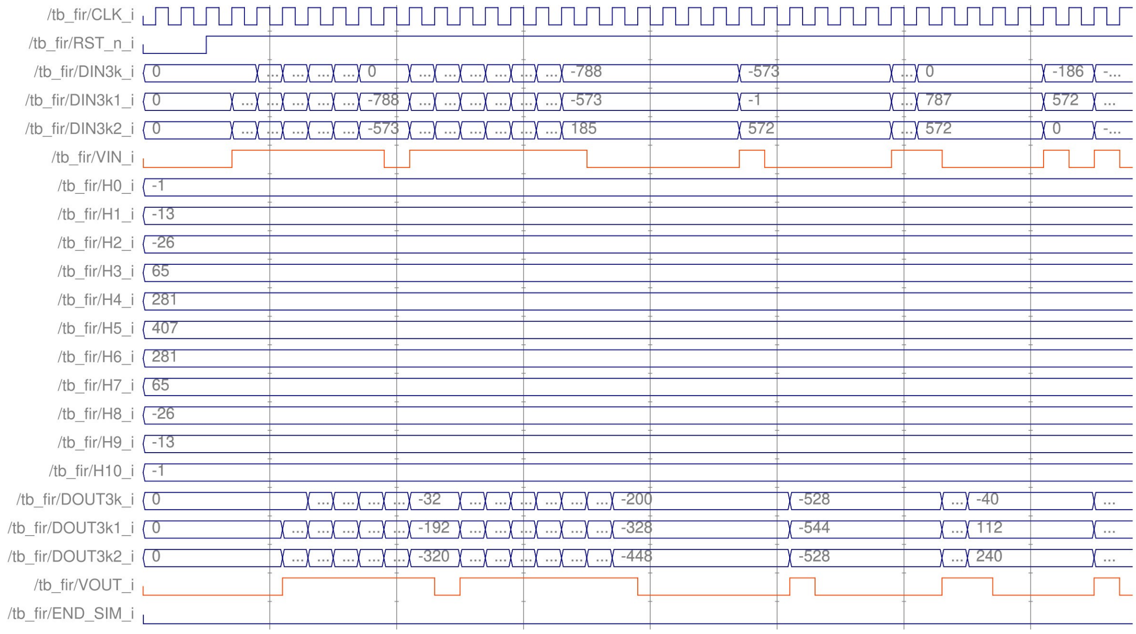

Figure 1.11: Simulation of the unfolded FIR

Frequency	Area	Power Consumption
$f_M = 274\ MHz$	$A = 17445.1\ \mu m^2$	$P = 4.07\ mW$
$f_M = 137\ MHz$	$A = 17037.1\ \mu m^2$	$P = 2.77\ mW$

Table 1.6: Unfolded FIR Synthesis results using backannotation and clock gating

1.3.3 Place & route

The Place & Route proceed as did in 1.2.4. By using the frequency mentioned in 1.3.2 we get positive slacks, *minimum slack* $= 0.061\ ns$. The circuit synthetized by `Innovus` produces no errorrs in the `QuestaSim` simulation used to generate the `.vcd` file, so it can be concluded that the architecture works. The simulation time, extracted using the procedure explained before, is $1,125\ \mu s$. Results of the Place & Route are shown in Table 1.7 and Figure 1.12. from 1.7 we can evaluate a power consumption reduction of ≈ 0.95, a little higher than the one evaluated in 1.2.3.

Frequency	Area	Power Consumption
$f_M/2 = 137\ MHz$	$A = 17808.7\ \mu m^2$	$P = 2.63\ mW$

Table 1.7: Place & Route results

1.3.4 Explanations, comparisons and comments

By applying the unfolding technique, the expected outcome is achieved - namely, an improvement in throughput and execution speed of the circuit, at the cost of a significant increase in power consumption. By comparing the simulation time measured in 1.2.4 and 1.3.3 we can see that the time required to complete the operation is reduced by ≈ 3, meaning the throughput is 3 times the original. Implementing the same logic by replicating, in a loop, the region performing multiplication and addition also leads to a considerable increase in the circuit area, which consequently becomes much larger.
The considerations made regarding the clock gating technique remain unchanged, as do those concerning the comparison between the results obtained using the different analysis methods.
The unfolded circuit consumes much more power than the folded implementation due to the increased hardwere,and also there is a noticable but small reduction in the efficency of powerconsumption optimization such as clock reduction and clock gating, as explained in 1.3.2.

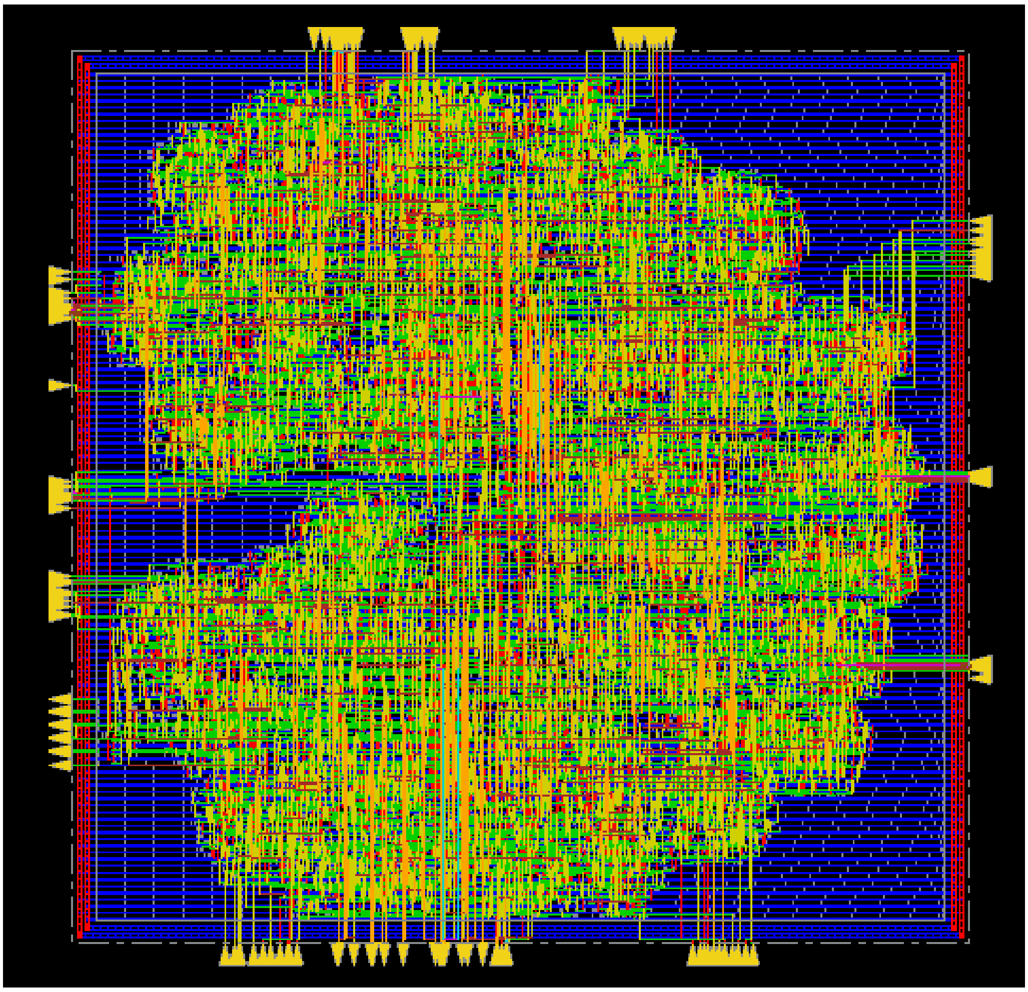

Figure 1.12: Unfolded Filter after Place & Route